The book Ramsey number $R(B_4, B_9) = 27$

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Formal proof in Lean 4 with SAT certificates

Abstract. The book B_n is n triangles sharing one edge. We show $R(B_4,B_9) = 27$. Before this work the value was known only to lie in {27, 28}. The lower bound comes from the strongly regular graph srg(26,10,3,4). For the upper bound, a degree-peeling argument reduces the claim to 27 SAT instances, each fixing the row of one vertex of given red degree. Each instance is refuted by a CaDiCaL LRAT certificate, 0.71 GB of trimmed proof in all. The whole argument is checked in Lean 4: Lean generates every CNF itself, a Lean proof connects each CNF to the definition of a book-free colouring, and Lean's verified LRAT checker accepts every certificate. The resulting theorem `LowerBound 26 4 9 ∧ UpperBound 27 4 9` depends only on Lean's standard axioms and the `native_decide` compiler-trust axiom. drat-trim's `lrat-check` also accepts all 27 certificates independently.

1. Introduction

For graphs G and H, the Ramsey number R(G,H) is the least N such that every red/blue colouring of the edges of K_N contains a red G or a blue H. Book Ramsey numbers were first studied systematically by Rousseau and Sheehan [RS78], and they have attracted renewed interest [CFW21, Wes24, LMPV24, DL26]. Books are also an ingredient in the recent exponential improvement for diagonal Ramsey numbers. Rousseau and Sheehan's counting bound gives $R(B_m,B_n) \le 2(m+n+1)$. For (m,n) = (4,9) this is 28. A strongly regular graph gives the lower bound 27 (Section 3). Radziszowski's dynamic survey [Rad, Table IXa] lists $R(B_4,B_9)$ as 27–28. The recent computational papers [LMPV24, Wes24] and the algebraic paper [DL26] do not treat this case.

Theorem 1. $R(B_4,B_9) = 27$.

The proof is computer-assisted. Section 2 fixes notation, Section 3 gives the lower bound, Section 4 gives the reduction, Section 5 the encoding, and Section 6 the certificates and their verification. Section 7 lists the controls we ran against encoding errors. Section 8 states exactly what has to be trusted.

2. Preliminaries

Identify a colouring of K_N with its red graph R on vertex set [N]. For a pair uv, write $c_R(u,v)$ for the number of common red neighbours and $c_B(u,v)$ for the number of common blue neighbours. A red B_m exists exactly when some red edge uv has $c_R(u,v) \ge m$; similarly a blue B_n exists exactly when some blue edge has $c_B(u,v) \ge n$. A colouring is *(m,n)-good* if it has neither. Then $R(B_m,B_n) > N$ if and only if some (m,n)-good colouring of K_N exists. The Lean development states this directly. `Statement.lean` is 23 lines, uses no library beyond core Lean, and is the only file a reader must audit to know what is proved:

```
def BookFree (red : Fin N → Fin N → Bool) (m n : Nat) : Prop :=
  ∀ u v, u ≠ v → (red u v = true  → redCodeg  red u v < m) ∧
                 (red u v = false → blueCodeg red u v < n)
def LowerBound N m n := ∃ red, Symm red ∧ BookFree red m n
def UpperBound N m n := ∀ red, Symm red → ¬ BookFree red m n
```

Here `redCodeg` and `blueCodeg` count, by list filtering over `Fin N`, the vertices w ∉ {u,v} adjacent to both u and v in red (respectively, in blue). So Theorem 1 is `LowerBound 26 4 9 ∧ UpperBound 27 4 9`.

3. The lower bound

Let G be any srg(26,10,3,4), for example a Paulus graph. Colour the edges of G red and all other pairs blue. A red edge has exactly $\lambda = 3 < 4$ common red neighbours. Take a blue pair uv. Exactly $\mu = 4$ vertices are red-adjacent to both, so $2k - \mu = 16$ vertices are red-adjacent to at least one of them. The common blue neighbourhood therefore has size $24 - 16 = 8 < 9$. Hence the colouring is (4,9)-good, and $R(B_4,B_9) \geq 27$. The Lean file `B49Witness.lean` contains an explicit such graph (130 edges, given in Appendix A) and checks `BookFree` by `native_decide`. The graph was found by the SAT pipeline of Section 5. Separately, a script independent of the CNF code confirms that it is 10-regular with $\lambda = 3$ and $\mu = 4$.

4. Reduction by degree peeling

Write d(v) for the red degree. For $0 \leq lo \leq d \leq hi \leq N-1$, let F(N,m,n,d,lo,hi) be the CNF of Section 5. It is satisfiable when an (m,n)-good colouring of K_N exists that has all red degrees in [lo,hi] and, after a suitable relabelling, has vertex 0 of red degree d.

Lemma 2 (row exclusion; Lean `row_excl_of_check`). Suppose F(N,m,n,d,lo,hi) is unsatisfiable. Then no (m,n)-good colouring of K_N with all red degrees in [lo,hi] has a vertex of red degree d.

Proof sketch. Suppose c is such a colouring and v has d(v) = d. Relabel so that v = 0 and its red neighbours are 1,…,d. Next, apply the permutation of {1,…,d} and the permutation of {d+1,…,N−1} that make the colouring lexicographically maximal among those fixing row 0. The result satisfies every clause of F, including the lex-leader clauses. Lean proves each step: relabelling preserves `Symm`, `BookFree` and red degrees, and a lex-maximal representative exists. ∎

Because [lo′,hi′] ⊆ [lo,hi], an exclusion proved for [lo,hi] also holds for every colouring whose degrees lie in [lo′,hi′]. Lemma 2 then lets us shrink the admissible degree interval one endpoint at a time (`peel_low`, `peel_high`). Let c be a (4,9)-good colouring of K_{27}. Its degrees start in [0,26], and the following refutations shrink that range to nothing:

Table 1. The peeling chain for N = 27, (m,n) = (4,9).

step	*instances (d; lo, hi)*	*degrees afterwards*
1	d = 26, 25, …, 11; [0,26]	[0,10]
2	d = 0, 1, …, 6; [0,26]	[7,10]
3	(7; 7,10)	[8,10]
4	(8; 8,10)	[9,10]
5	(10; 9,10)	[9,9]
6	(9; 9,9)	∅

For step 6, a 9-regular graph on 27 vertices cannot exist by the handshake lemma. The certificate refutes this case directly anyway; the parity argument is not used in Lean.

Remark: why the instances are small

The following count explains why the instances are easy, and the paper's other numbers do not depend on it. By Goodman's identity, the number of monochromatic triangles is $T = C(27,3) - \tfrac{1}{2}\Sigma_v\, d(v)(26 - d(v))$. In a (4,9)-good colouring each red edge lies in at most 3 red triangles and each blue edge in at most 8 blue triangles. With r red edges this gives $T \le r + 8(351 - r)/3$. Combining the two, $\Sigma_v\, f(d(v)) \ge 1989$, where $f(d) = 73d/6 - d^2/2$. The maximum of f is f(12) = 74, so the total deficit $\Sigma(74 - f(d(v)))$ is at most 9. The deficit is 0 at d = 12, 1/3 at d = 13, 2/3 at d = 11 and 7/3 at d = 10. Counting alone therefore forces nearly all degrees into {11,12,13}, and it excludes every colouring with all degrees in [8,10]. Rousseau and Sheehan's bound of 28 is the same count at N = 28. The substantive content of the certificates is that no vertex can have red degree 11, 12 or 13. Accordingly, the d = 11 certificate is the largest (Table 2).

5. Encoding

Each pair {i,j} gets one variable, true meaning red. Further Tseitin variables t^R_{ijk} and t^B_{ijk} are shared by the three pairs of each triple and mark red and blue triangles. For every pair {i,j}, a sequential counter [Sin05] over the N−2 red triangle variables, guarded by x_{ij}, enforces at most m−1 of them. A second counter over the blue triangle variables, guarded by $\neg x_{ij}$, enforces at most n−1. Row 0 is fixed by unit clauses (x_{0j} true exactly for $1 \le j \le d$). Sequential counters bound every red degree to [lo,hi]. Finally, lex-leader clauses order consecutive rows within the blocks {1,…,d} and {d+1,…,N−1}. For N = 27 the instances have 122,958–124,362 variables and 264,560–267,260 clauses. The CNF is defined in Lean (`RowCNF.lean`, function `rcnf`) as a pure function. The DIMACS files given to the solver are printed by a compiled Lean executable from that same definition, so the formula Lean checks is exactly the formula that was solved.

6. Certificates and verification

CaDiCaL 3.0.1 produced each refutation as binary LRAT. Lean's `lrattrim` trimmed it and gzip compressed it. Each certificate is embedded in a Lean module as Base64 text. Lean decodes it with a DEFLATE decoder written in pure Lean, parses it, and checks it against `rcnf 27 4 9 d lo hi` with the

LRAT checker of Lean's standard library (`Std.Tactic.BVDecide.LRAT.check`), whose soundness is proved in Lean. The decoder's correctness is not part of the trusted base: a faulty decoder could only produce a certificate the checker rejects, never a false theorem. The checking computation runs under `native_decide`.

Table 2. The 27 certificates (bytes of trimmed text LRAT; Lean module build time).

d	*lo*	*hi*	*trimmed LRAT*	*gzip*	*Lean (s)*	*d*	*lo*	*hi*	*trimmed LRAT*	*gzip*	*Lean (s)*
26	0	26	15,020	3,921	281	13	0	26	1,394,189	300,706	286
25	0	26	20,848	5,294	282	12	0	26	5,386,776	1,096,334	297
24	0	26	44,393	9,812	272	11	0	26	356,718,357	71,350,240	1,116
23	0	26	21,186	5,403	273	0	0	26	10,543	2,666	304
22	0	26	26,037	7,009	274	1	0	26	125,938	26,584	309
21	0	26	23,550	6,032	276	2	0	26	4,295,360	891,272	292
20	0	26	21,170	5,754	287	3	0	26	4,912,499	986,627	292
19	0	26	31,831	8,054	287	4	0	26	6,053,388	1,235,969	264
18	0	26	19,461	4,860	298	5	0	26	43,798,210	10,385,704	353
17	0	26	15,519	4,216	295	6	0	26	267,533,942	63,315,794	875
16	0	26	68,762	17,369	287	7	7	10	4,331,226	901,579	641
15	0	26	170,301	46,411	288	8	8	10	12,776,008	2,319,164	645
14	0	26	432,138	93,729	280	10	9	10	2,855,260	588,973	606
						9	9	9	739,140	93,373	556
total			711,841,052 B trimmed · 153,712,849 B gzip			10,516 s of Lean build in total; final assembly 2.2 s					

The final module `R49Final.lean` chains the 27 exclusions as in Table 1. Lean reports:

```
'BookRamsey.R49.R_B4_B9_eq' depends on axioms: [propext, Classical.choice, Quot.sound,
  BookRamsey.b49_lower._native.native_decide.ax_1_1,
  BookRamsey.R49.check_d00_lo0_hi26._native.native_decide.ax_1_1, … (27 such check
axioms)]
```

No `sorry` or user-declared axiom occurs. Each `native_decide` axiom says that one specific closed Boolean computation, an LRAT check or the witness check, evaluates to `true` when compiled.

Independent check

All 27 trimmed certificates were also checked with `lrat-check` from drat-trim [WHH14], against the same DIMACS files. This checker shares no code with CaDiCaL or Lean, and it accepted all 27.

7. Controls against encoding errors

The instances are much easier than the neighbouring open case $R(B_4,B_7)$ (Section 9), so we ran the following controls. (i) *Satisfiable control at N = 26:* the same generator, with (d;lo,hi) = (10;10,10), returns SAT in 4.7 s. The model is an srg(26,10,3,4) and passes an independent codegree check. Since the Section 4 remark makes 10-regular colourings Goodman-tight at N = 26, this tests the encoding exactly at the extremal configuration. (ii) *Independent generator:* a separately written Python encoder

with the same semantics but different code agrees on the decisive cases at N = 27, (8;8,10), (10;9,10) and (9;9,9), all UNSAT, and on the N = 26 SAT control. (iii) *Known values:* the pipeline reproduces $R(B_3,B_5) = 17$, $R(B_2,B_9) = 22$ and $R(B_4,B_6) = 22$, with SAT on the corresponding critical sizes. $R(B_4,B_6) \leq 22$ is formalised end to end in the same Lean framework. Controls (i)–(iii) are evidence for the encoding, not part of the proof. The proof rests on Lemma 2, which Lean proves against `Statement.lean`.

8. Trusted base

To accept Theorem 1 from the artefact, one must trust: the Lean 4 kernel (v4.34.1); the faithfulness of `Statement.lean` to the mathematical definition (Section 2); and the Lean compiler and runtime used by `native_decide` for 28 closed computations. One need not trust CaDiCaL, the CNF printer, the DEFLATE decoder, the cube or peeling scripts, or any Python code. The `native_decide` trust could be removed with a kernel-only `decide` on smaller certificates, but that is impractical at 0.7 GB. Section 6 records the independent drat-trim check, which gives assurance that does not depend on Lean's compiler.

9. Related work and outlook

The SAT approach to book Ramsey numbers follows [Wes24], and the flag-algebra bounds are from [LMPV24]. Combining row fixing and degree peeling with a Lean-checked bridge to the definition is, as far as we know, new for these problems. The same pipeline is being applied to $R(B_4,B_7) \in \{22,23\}$. There every case of the analogous chain has been refuted, and all 347 certificates pass drat-trim's checker, but the Lean assembly (tens of gigabytes of trimmed proof) was still building when this paper was written. We make no claim about $R(B_4,B_7)$ here. All negative attempts, including circulant and block-circulant searches, Goodman cuts and monolithic solving, are logged in the accompanying ledger.

Appendix A. The witness

An srg(26,10,3,4) on vertices 0–25 as an edge list. The same list appears in `B49Witness.lean`:

```
130 edges: 0–1 0–2 0–3 0–4 0–5 0–6 0–7 0–8 0–9 0–10 1–2 1–3 1–4 1–11 1–12 1–13 1–14 1–15
1–16 2–3 2–5 2–11 2–17 2–18 2–19 2–20 2–21 3–6 3–12 3–17 3–22 3–23 3–24 3–25 4–7 4–8 4–13
4–14 4–18 4–19 4–22 4–23 5–7 5–9 5–15 5–16 5–18 5–20 5–22 5–24 6–9 6–10 6–13 6–15 6–18 6–
21 6–23 6–25 7–10 7–13 7–16 7–17 7–19 7–24 7–25 8–9 8–10 8–12 8–14 8–17 8–20 8–21 8–22 9–
11 9–14 9–16 9–21 9–23 9–24 10–11 10–12 10–15 10–19 10–20 10–25 11–14 11–16 11–19 11–20
11–23 11–25 12–15 12–16 12–19 12–21 12–22 12–24 13–15 13–16 13–17 13–20 13–21 13–23 14–15
14–17 14–18 14–24 14–25 15–18 15–20 15–24 16–21 16–22 16–25 17–20 17–21 17–24 17–25 18–19
18–21 18–22 18–25 19–21 19–23 19–24 20–22 20–23 22–23 22–25 23–24
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